INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, First Semester Examination, November - 2024

Subject Code: BCACC103

Subject Name: Discrete Structure

Full Marks: 30 Duration: 1 Hour

Part - A
Attempt 5 questions
Each question carries 2 marks (2 × 5)

1. Define Simple Graph with proper example. 2

or

Define regular graph with proper example. 2

2. Explain the concept of an isolated vertex with proper examples. 2

or

Explain the properties of a complete graph. 2

3. Draw a graph explaining the existence of adjacent edges. 2

or

Draw a graph which is not simple, having two isolated vertices. 2

4. Draw a binary tree with four edges. 2

or

Draw a tree with five vertices and total degree 8. 2

5. How many edges a complete graph has with 10 vertices? 2

or

The sum of the degrees of all the vertices of a graph is 40. Find the number of edges of the graph. 2

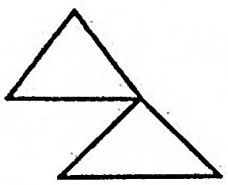
Part - B
Attempt 2 questions
Each question carries 5 marks (5 × 2)

6. Draw the graph whose incidence matrix is $\begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$ 5

or

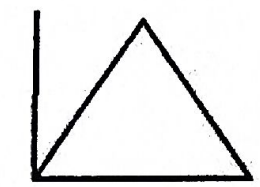
Draw the graph whose incidence matrix is $\begin{bmatrix} 0 & 0 & 1 & -1 & 1 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix}$ 5

7. Determine the adjacency matrix of the following graph : 5



or

Determine the adjacency matrix of the following graph : 5

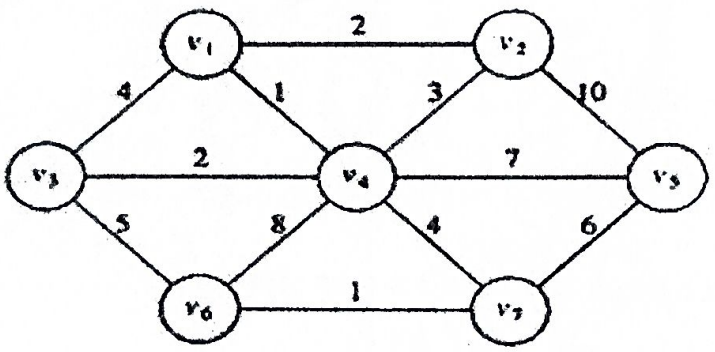


Part - C

Attempt 1 question

Each question carries 10 marks (10 × 1)

8. Using Kruskal's Algorithm find the minimal spanning tree and it's total weight from the following graph **10**



or

Using Prim's Algorithm find the minimal spanning tree and it's total weight from the following graph **10**

